

Kareem Elkhatib

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EDUCATION

University of Georgia

Bachelor of Science in Electrical Engineering

GPA: 3.55

Expected Graduation: May 2028

EXPERIENCE

GaSuitWarehouse.com | Menswear retail business

Jan 2025 - Present

Growth Operator

Gainesville, GA

- Ran a local SEO strategy that **ranked the store top 3 on Google** for high-intent "suit store near me" searches, outranking established competitors and driving **~60% of store traffic** from organic search.
- Built the company website with an integrated booking funnel plus SMS and email reminder automations, cutting no-shows and removing manual follow-up.
- Optimized the booking flow's conversion rate to turn visitors into booked appointments, adding **~\$6,000 in monthly recurring revenue**.

PROJECTS

Power-Rail Capacitor Health Monitor

April 2026

- Developed a diagnostic system that identifies a failing capacitor **before it takes down the device** it powers — a leading cause of downtime that costs data centers **\$100K-\$500K per hour**.
- Programmed the system to read the power signal **once per second** and classify the part as healthy, borderline, or bad, achieving **100% accuracy across 4 swapped capacitors**.
- Verified the results were reliable by testing across **10 Hz - 100 kHz** and confirming the measurements matched circuit theory predictions exactly.

Op-Amp Low-Pass Filter

Nov 2025

- Assembled a **TL081 active low-pass filter** on a breadboard and powered it from ± 5 V rails using AD3
- Derived the filter's **transfer function and cutoff frequency** from the circuit model, then verified expected behavior through bench testing.
- Used the AD3 network analyzer to confirm the frequency response and noise-reduction behavior.

External Camera Power System

June 2025

- Engineered an external camera power system by matching OEM battery voltage requirements and integrating a dummy-battery + V-mount setup, **increasing runtime ~20×** and enabling full-day shoots.
- Validated power stability under load with extended-duration tests, confirming reliable field operation.
- Improved on-location power reliability and reduced setup/reset time by 20% per shoot.

SKILLS

- AI Automation | Agentic Workflow Design | AI-Enabled Software Development | Prompt Engineering
- Problem Solving | Analytical | MultiSim | Circuit Analysis | Excel
- Languages | Python | MATLAB | RLAB | C++

CERTIFICATIONS

- **Databricks | Generative AI Fundamentals (Academy Accreditation)** — Core generative AI concepts and applied use.
- **OpenAI | Agents and Workflows** — Building agents through context, defined outputs, boundaries, and review loops.